VoiceLog – A Voice-Powered Smart Expense Tracking Application

Project Name: VoiceLog
Domain: Artificial Intelligence | Voice Technology | Personal Finance

Problem Statement

Manually recording daily expenses can be time-consuming and inconvenient. Users may forget to log small transactions, especially when they are busy. Traditional expense trackers also require users to enter each expense individually, making the process less convenient.

Proposed Solution

VoiceLog is a voice-powered expense tracking application that allows users to record multiple expenses simply by speaking. It uses Gnani AI's speech-to-text technology to convert voice input into text and processes the recognized information to identify expense items, amounts, and categories.

Key Features

- **Voice-Based Expense Entry:** Record expenses using natural speech instead of manual typing.
- **Multiple Expense Extraction:** Identify and process multiple expenses from a single voice recording.
- **Automatic Categorization:** Organize expenses into relevant categories such as Food & Dining and Travel.
- **Multilingual and Mixed-Language Input:** Support practical expense logging through English and Tanglish voice input.
- **Expense History:** Store and display recorded expenses for easy tracking.

Technology Stack

- **Speech Recognition:** Gnani AI Prisma Speech-to-Text API
- **Backend:** Python and Flask
- **Desktop Application:** Electron
- **Frontend:** HTML, CSS, and JavaScript
- **Data Storage:** JSON-based local storage

Implementation Approach

The user records an audio message through the application. The audio is processed and sent to the Gnani AI speech-to-text API. The returned transcript is analyzed to extract individual expense descriptions and their corresponding amounts. The application then assigns suitable categories and stores the expense details for display in the expense history.

Expected Impact

VoiceLog aims to make everyday expense tracking faster, simpler, and more accessible by reducing manual data entry. It provides a convenient way to record multiple transactions through a single voice interaction and helps users maintain a record of their daily spending.

Conclusion

VoiceLog combines speech recognition and automated expense processing to simplify personal expense management. By making expense recording as simple as speaking, it aims to encourage users to track their spending more consistently.